Newton's second law: vary mass

LEARNING TARGET · S3 I can measure how acceleration changes with mass at constant force.

A · KEY VOCABULARY

Fair test	Change one thing at a time.
Changed variable	Total system mass: cart + added masses + hanger.
Controls	Hanging mass, track, start point, release (no push).
Evidence	The slope of each velocity–time graph is the acceleration.

B · KEY EQUATION + PATTERN

$$a \propto 1 \div m$$

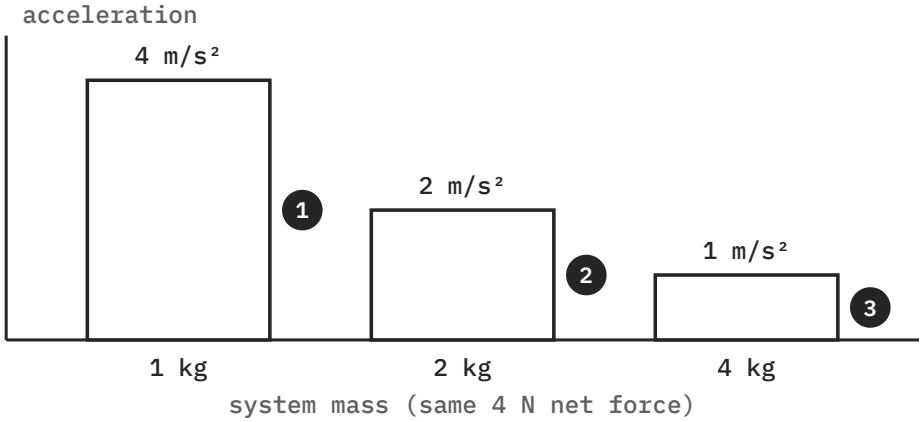
$$m \times a \approx \text{constant}$$

$$a = \Delta v \div \Delta t \text{ (slope)}$$

mass	kilograms (kg)
a	m/s²

C · COLOR IT · DOUBLE THE MASS

Illustrative values — not your data.



- ☐ **1 · GREEN** — 1 kg system (dark)
- ☐ **2 · GREEN** — 2 kg system (medium)
- ☐ **3 · GREEN** — 4 kg system (light)

D · WORKED EXAMPLE

In one trial, velocity rises from 0 to 4 m/s in 2 s. Find the acceleration.

GIVEN	EQUATION	WORK	ANSWER
$v_i = 0$ m/s $v_f = 4$ m/s $\Delta t = 2$ s	$a = \Delta v \div \Delta t$	$\Delta v = 4 - 0 = 4$ m/s $a = 4 \div 2$	**2 m/s²** use your own measured graph

E · THE 2026-XJ THREAD

DAY 11 OF 22 · NEW TOOL: MASS

A massive object is hard to nudge. The bigger XJ's mass, the smaller the acceleration any push or pull gives it, which matters for any plan to deflect it.

Unit question: How fast is it moving, and where will it arrive?

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else